Quantitative fluid dynamics with the extended jet-flow facility

1 Objective of the experiment

The aim of this experiment is twofold. First, students observe how a controlled stream of air behaves at different fan speeds and how an external sound source can disturb that stream. Second, and new in this version of the setup, students obtain a numerical measurement of the air velocity inside the facility by reading the differential pressure across the stagnation chamber and applying Bernoulli's equation. The facility therefore moves from a qualitative demonstration toward a small quantitative laboratory experiment.

By the end of the session the student should be able to relate a visible flow pattern to a measured pressure value, recognise the transition between laminar and turbulent regimes, and describe how acoustic forcing modifies a jet.

2 Theoretical fundamentals

2.1 Fluid dynamics

Fluid dynamics is the branch of physics that describes the motion of gases and liquids and the forces acting on them. It underpins applications as varied as aircraft design, weather prediction, blood flow modelling, and ventilation. In the laboratory it is usually approached through simplified geometries that isolate one phenomenon at a time, which is the role this facility plays.

2.2 The jet-flow facility

The setup used here is a single-jet wind facility. A fan draws ambient air through an intake, the airflow is conditioned by a diffuser followed by a honeycomb structure, and the resulting stream exits through a nozzle. The diffuser slows the flow so that the honeycomb can straighten it into parallel filaments, producing the laminar profile required for clean visualisation and measurement.

2.3 Smoke visualisation

Air on its own is invisible, so it has to be marked with a tracer to become observable. A small vaporiser produces a non-toxic glycol mist which is light enough to follow the local velocity field with very little lag. Once the mist is entrained in the airstream, the eye can clearly distinguish smooth laminar filaments from chaotic turbulent eddies.

2.4 Acoustic forcing

Sound consists of small periodic pressure fluctuations travelling through the air. When these fluctuations reach a jet, they perturb the shear layer at the edge of the stream and can trigger or suppress the natural instabilities that would otherwise grow further downstream. The result is a jet whose behaviour depends not only on the fan speed but also on the frequency and amplitude of the imposed sound. This is referred to as acoustic forcing.

2.5 Pressure measurement and Bernoulli's principle

The extension added to the facility lets the user read the air velocity rather than estimate it visually. A differential pressure sensor sits between the stagnation chamber inside the facility and the ambient air outside. The difference between these two pressures, called the dynamic pressure, depends directly on the flow speed.

For an incompressible airflow at low speeds, Bernoulli's equation relates the dynamic pressure ΔP to the velocity v through:

$$\mathbf{v = \sqrt{(2 \cdot \Delta P / \rho)}}$$

where ρ is the density of air, taken as 1.225 kg/m^3 at standard room conditions. The microcontroller performs this calculation internally and displays both the raw pressure and the resulting velocity on the second LCD, so no manual computation is required during the session.

A few caveats are worth keeping in mind. Bernoulli's equation assumes a steady, frictionless, incompressible flow along a streamline. In a real jet these assumptions break down slightly, so the displayed velocity should be treated as a good estimate rather than an exact value. The sensor itself also has a small zero offset of roughly 200 Pa which appears as a non-zero reading even when the fan is off. Subtract this offset before drawing conclusions.

3 Work instructions

3.1 Preparing the facility

Verify before powering anything that the fan speed potentiometer (-SGA1) and the speaker volume control (-SGA2) are both turned fully counter-clockwise to the zero position. Plug the 12 V power supply (-TBA1) into a wall outlet and connect its barrel jack to the control panel inlet (-XDB1). Plug the amplifier (-TFA1) into a wall outlet using its dedicated mains cable (-XDB2). Press the power switch (-SJA1) on the control panel. The yellow LCD (-PHA1) should now show the current fan speed at 0 percent and the green LCD (-PHA2) should show a pressure reading near zero and a velocity near zero. Once both displays are active, the fan can be brought up by turning -SGA1 clockwise.

3.2 Reading the displays

Two LCD screens are mounted on the control panel. The yellow one shows the current fan speed as a percentage of full power, which corresponds to the position of the speed potentiometer. The green one shows two values: the differential pressure across the stagnation chamber in pascals, and the air velocity in metres per second derived from that pressure. The pressure value may oscillate by a few pascals as the sensor picks up small acoustic disturbances. This is normal and indicates that the sensor is working.

3.3 Smoke visualisation module

Slot the smoke machine (-HPB1) into the nozzle adapter (-NAD1) at the outlet end of the facility. Press the wireless smoke button (-SJB1) to release a puff of mist into the airflow. Vary the fan speed

(-SGA1) across its range and observe how the visible plume changes shape. At low speeds the smoke should travel in clean parallel layers. As the speed rises, the smoke breaks up into irregular swirling structures. Note your observations together with the corresponding pressure and velocity readings from the green LCD. Repeat the exercise with the smoke machine moved into the fan-side adapter (-NAD2) and compare the patterns at the same flow speeds.

3.4 Acoustic forcing module

For this part of the experiment the fan-side smoke adapter (-NAD2) should remain in place while the nozzle adapter (-NAD1) is removed. Attach the acoustic adapter (-XMB1) to the nozzle. Connect the audio cable (-XDB3) to the amplifier (-TFA1) on one end and to a phone or audio player on the other end. Toggle the speaker switch (-SJA2) to enable the speaker. Begin smoke production with (-SJB1) and play music while varying the volume with the speaker control (-SGA2). Try tracks of different genres, particularly those with strong low-frequency content, and document how the visible jet responds. Pay attention to any changes in the green LCD pressure reading during loud low-frequency passages.

3.5 Quantitative measurement procedure

This section is new compared to the previous version of the experiment and produces the numerical data needed for the analysis questions below.

Step 1. With the fan switched off, record the resting pressure value displayed on the green LCD. This is the sensor zero offset. Typical values fall in the range of 200 Pa to 230 Pa.

Step 2. Set the fan to 10 percent using -SGA1 and wait approximately ten seconds for the readings to stabilise. Note both the pressure and the velocity from the green LCD, together with the fan speed setting.

Step 3. Repeat at 25, 50, 75 and 100 percent fan speed. Five well spaced data points are sufficient.

Step 4. For each measurement, subtract the offset from Step 1 from the raw pressure reading. The corrected pressure is the value to use for any velocity calculation done by hand.

Step 5. Plot the corrected pressure on the vertical axis and the fan speed percentage on the horizontal axis. Then plot velocity against fan speed on a second graph. Compare the shapes of the two curves and connect them to the underlying physics in your analysis.

3.6 Analysis questions

Answer the following questions in your written submission. Combine visual observations from the smoke modules with numerical results from the pressure measurements where possible.

1) How does the visible behaviour of the airflow change as the fan speed increases?

2) At what approximate fan speed does the jet appear to transition from laminar to turbulent? Does this match the pressure or velocity reading at which it happens?

3) Which fluid dynamics principles explain the laminar to turbulent transition?

4) How does smoke visualisation help to interpret flow phenomena that would otherwise be invisible?

5) Using your recorded data, plot the measured velocity against fan speed percentage. Is the relationship linear, quadratic, or something else? Explain why this shape is expected.

6) Using your recorded data, plot the corrected pressure against fan speed percentage. How does this shape relate to the velocity curve?

7) Estimate the Reynolds number at maximum fan speed using a characteristic length equal to the nozzle diameter and a kinematic viscosity of 1.5×10^{-5} m^2/s. Does the value support your visual observation of laminar versus turbulent flow?

8) What effect does the acoustic forcing have on the visible airflow at different volumes and frequencies?

9) Through which physical mechanisms can sound waves modify the structure of a jet?

10) Discuss the main sources of measurement uncertainty in the pressure and velocity readings, and how the experiment could be improved.

11) Suggest a practical engineering application that could benefit from acoustic control of an airflow.

4 Material list

Component references follow the IEC 81346-2 standard. Items marked with an asterisk were added or modified in the current revision of the facility.

Reference	Name	Comment
-BPB1*	Pressure sensor	ELVH-M250D-HRRJ-I-N3A5; +/-250 mbar diff., I2C 0x38
-CAA1	Capacitor	1nF; 25V
-GQB1	Jet-flow facility	Inherited from previous project group
-HMC1*	I2C multiplexer hub	TCA9548A based; 8 Grove ports; addr 0x70
-HPB1	Smoke machine	Small vaporiser
-KEB1	Microcontroller	ESP32 NodeMCU-32S
-MAA1	Fan motor	12VDC; 1.3A; 15.6W
-NAD1	Smoke adapter nozzle	3D-printed; two parts
-NAD2	Smoke adapter fan	3D-printed
-PFA1	Power LED	3V; 20mA; green
-PHA1	Fan speed display	Grove RGB LCD V5.0; yellow backlight; I2C hub ch 0
-PHA2*	Pressure display	Grove RGB LCD V5.0; green backlight; I2C hub ch 1
-PJA1	Speaker	5 inch woofer; 4 Ohm
-RCA1	Limiting resistor	15 Ohm; series with -PFA1
-SGA1	Fan speed control	10 kOhm potentiometer
-SGA2	Speaker volume level	Rotary knob on amplifier
-SJA1	Power on/off	Momentary toggle switch
-SJA2	Speaker on/off	Toggle switch
-SJB1	Smoke trigger	Push button; wireless remote
-TAB1	DC-DC converter	Mean Well SD-15A-05; 12VDC to 5VDC; 3A
-TBA1	Power supply	12VDC; 3A external brick
-TFA1	Amplifier	50W; 4 Ohm; t.amp PM40C
-UCA1	Control panel	3D-printed box; laser-cut top
-XDB1	Control panel inlet	Barrel socket; 12VDC; 5A
-XDB2	Cold device plug	1.5m mains cable for amplifier
-XDB3	Audio plug	2.5m audio cable
-XMB1	Acoustic adapter	3D-printed with glued tubes

4.1 Note on the pressure sensor pinout

The ELVH-M250D-HRRJ-I-N3A5 uses the HRRJ surface-mount J-lead package. The Pin Code 1 table in the standard ELVH datasheet places SCL on pin 5, but on this particular variant SCL is on pin

4 and pin 5 is unused. The correct connections, verified empirically and confirmed by the supplier, are listed below.

Sensor pin	Function	Wire colour	Connects to
1	GND	Black	Hub channel 2 GND
2	Vs (supply, 3.3V)	Red	Hub channel 2 VCC
3	SDA	White	Hub channel 2 SDA
4	SCL	Yellow	Hub channel 2 SCL
5 to 8	Not connected	n/a	Leave open

5 The experiment's set-up

The complete facility consists of the original jet-flow tube with its fan, smoke and acoustic modules, together with the extended control panel that now hosts two LCD screens and the pressure sensor measurement chain.

Key signal flow on the electronics side:

- The ESP32 microcontroller (-KEB1) runs the firmware and exposes a single I2C bus on its pins D21 (SDA) and D22 (SCL).
- This single I2C bus is fanned out to multiple downstream devices through the TCA9548A multiplexer hub (-HMC1).
- Channel 0 of the hub drives the fan speed LCD (-PHA1).
- Channel 1 of the hub drives the pressure and velocity LCD (-PHA2).
- Channel 2 of the hub connects to the pressure sensor (-BPB1) inside the stagnation chamber.
- Channels 3 to 7 are kept free for future expansion such as adding a temperature sensor or further user controls.

On the power side, the external 12 V supply (-TBA1) feeds both the fan and the on-board DC-DC converter (-TAB1), which steps the voltage down to 5 V for the ESP32 and the rest of the low-voltage electronics. The amplifier (-TFA1) has its own mains supply via a cold-device plug and drives the speaker (-PJA1) independently of the microcontroller, just as in the original setup.

A photograph of the assembled control panel and a wiring schematic are to be inserted here in the final document. The schematic should show the ESP32, the I2C multiplexer hub, both LCD displays, the pressure sensor, the fan motor, the LED, the toggle switches, the potentiometer and the DC-DC converter.

This worksheet is an updated revision of the original document produced by Group 2 (UFO – SSJFF): Finja Dittmann, Jelte Hoekstra, Robin Lind and Noud van der Meulen. The current revision was prepared by Group 5: Jense Cool, Juan Trigueros, Alec Vroon and Siiri Vainopaa, and extends the original facility with quantitative pressure and velocity measurement.